Lecture 5: Fundamentals of Observational Cosmology (I)

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Abstract

This lecture begins the observational side of the course using Type Ia supernovae as a concrete example. We review why supernovae can be treated as standardizable candles, how luminosity distance depends on cosmology, what the basic supernova dataset looks like, and how one turns redshift-distance measurements into cosmological parameter constraints. A compact Fortran example is included after the physics discussion rather than before it.

Learning goals

After this lecture, students should be able to:

- explain why Type Ia supernovae are useful distance indicators;
- define the distance modulus and luminosity distance;
- write the FLRW expression for $D_L(z)$ in curved and flat cosmologies;
- understand the basic structure of a supernova dataset and a χ^2 fit;
- implement a simple numerical evaluation of $D_L(z)$.

1 Why Type Ia supernovae matter

Type Ia supernovae (SNe Ia) were the first direct observational probe to show that the cosmic expansion is accelerating. They are not perfect standard candles, but after empirical light-curve corrections they become good *standardizable* candles. The central idea is simple:

- spectroscopy gives us the redshift z;
- calibrated photometry gives us the apparent brightness;
- comparing intrinsic and observed brightness gives the luminosity distance.

The 1998–1999 supernova results by the Riess team and the Perlmutter team are landmark papers in observational cosmology [1, 2].

2 Distance modulus

Astronomers often work with magnitudes instead of fluxes. The distance modulus is

$$\mu \equiv m - M = 5\log_{10}\left(\frac{D_L}{\mathrm{Mpc}}\right) + 25,$$

where m is apparent magnitude, M is absolute magnitude, and D_L is the luminosity distance in Mpc.

This equation is one of the most practical formulas in observational cosmology. Given a cosmological model, we can compute $D_L(z)$ and therefore predict $\mu(z)$.

3 Luminosity distance in FLRW cosmology

Define the dimensionless expansion function

$$E(z) \equiv \frac{H(z)}{H_0}.$$

For a matter-plus-curvature-plus-cosmological-constant model,

$$E(z) = \sqrt{\Omega_{\rm m}(1+z)^3 + \Omega_\Lambda + \Omega_k(1+z)^2}, \qquad \Omega_k = 1 - \Omega_{\rm m} - \Omega_\Lambda.$$

The line-of-sight comoving distance is

$$D_C(z) = \frac{c}{H_0}\int_0^z \frac{{\rm d}z'}{E(z')}.$$

The transverse comoving distance is

$$D_M(z) = \begin{cases} \frac{c}{H_0\sqrt{\Omega_k}}\sinh\left(\sqrt{\Omega_k}\,\frac{H_0 D_C}{c}\right), & \Omega_k > 0, \\ D_C, & \Omega_k = 0, \\ \frac{c}{H_0\sqrt{-\Omega_k}}\sin\left(\sqrt{-\Omega_k}\,\frac{H_0 D_C}{c}\right), & \Omega_k < 0. \end{cases}$$

Finally,

$$D_L(z) = (1+z)D_M(z).$$

Flat-universe special case. If $\Omega_k = 0$, then

$$D_L(z) = (1+z)\frac{c}{H_0}\int_0^z \frac{{\rm d}z'}{E(z')}.$$

This is the expression most often used in simple classroom exercises.

4 What the data look like

A basic supernova dataset contains:

- redshifts z_i;
- measured distance moduli μ_i;
- an error model, often summarized by a covariance matrix C_{ij}.

In a realistic analysis, one must also account for calibration, intrinsic scatter, host-galaxy effects, and nuisance parameters such as the absolute magnitude. For an introductory lecture, however, it is reasonable to start with the idealized relation

$$\chi^2(\boldsymbol{\theta}) = \sum_{i,j}\left[\mu_i - \mu_{\rm th}(z_i;\boldsymbol{\theta})\right](C^{-1})_{ij}\left[\mu_j - \mu_{\rm th}(z_j;\boldsymbol{\theta})\right].$$

5 Why supernovae can reveal acceleration

At low redshift, all reasonable cosmological models look similar because the Hubble law dominates. The discriminating power comes from how the distance-redshift relation bends at intermediate redshift.

A decelerating Universe and an accelerating Universe predict systematically different values of $D_L(z)$ and therefore different values of $\mu(z)$. The supernova Hubble diagram thus becomes a direct test of the expansion history.

6 A compact Fortran example

The original notes contained a long code listing before enough physical explanation. A better order is:

1. understand the distance formulas;
2. translate each formula into a function;
3. test the code at a few known redshifts;
4. only then connect it to a real dataset.

The minimal structure of a Fortran program is shown below.

```
module precision
  implicit none
  integer, parameter :: dp = kind(1.d0)
end module precision

module cosmology
  use precision
  implicit none
  real(dp) :: om = 0.3_dp, ol = 0.7_dp, H0 = 70._dp
  real(dp), parameter :: c = 299792.458_dp
contains
  real(dp) function E_of_z(z)
    real(dp), intent(in) :: z
    real(dp) :: ok
    ok = 1._dp - om - ol
    E_of_z = sqrt(om*(1._dp+z)**3 + ol + ok*(1._dp+z)**2)
  end function E_of_z
end module cosmology
```

In practice you would then add:

- a numerical integral for $D_C(z)$;
- a wrapper to convert D_C to D_L;
- a function that returns $\mu(z)$;
- a loop over data points to accumulate χ^2.

7 Recommended computational workflow

For undergraduate work, the following workflow is usually more educational than pasting in a long code block:

1. verify $E(z)$ by hand for a few redshifts;
2. implement the integral for $D_C(z)$;
3. compare flat and curved models;
4. plot $\mu(z)$ for several cosmologies;
5. only then fit the data.

8 Summary

This lecture introduced the supernova Hubble diagram as a clean entry point into observational cosmology. The essential formulas are sufficient to reproduce the classic discovery-level supernova analysis in simplified form [1, 2]. They can also be implemented numerically using standard integration routines such as those discussed in [3]. The essential formulas are:

$$\mu = 5\log_{10}(D_L/\mathrm{Mpc}) + 25,$$

$$D_C(z) = \frac{c}{H_0}\int_0^z \frac{\mathrm{d}z'}{E(z')}, \qquad D_L(z) = (1+z)D_M(z).$$

These equations turn an observed redshift and brightness into a test of cosmic expansion.

Suggested reading

- Riess et al. (1998) and Perlmutter et al. (1999) [1, 2].
- Numerical Recipes for practical numerical integration background [3].
- Dodelson and Schmidt, sections on distance measures and supernova cosmology.

Homework

1. **Low-redshift limit.** Show that for sufficiently small z,

$$D_L(z) \approx \frac{cz}{H_0}.$$

   Use this to derive the leading small-z form of the distance modulus.

2. **Numerical calculation.** Assume a flat ΛCDM model with $\Omega_\mathrm{m} = 0.3$, $\Omega_\Lambda = 0.7$, and $H_0 = 70\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. Evaluate $D_L(z)$ and $\mu(z)$ at $z = 0.5$ and $z = 1.0$.

3. **Coding exercise.** Write a short program in Fortran, Julia, or Python that computes $\mu(z)$ on a grid from $z = 0$ to 1.5 for three models: $(\Omega_\mathrm{m}, \Omega_\Lambda) = (1, 0)$, $(0.3, 0.7)$, and $(0.3, 0)$. Plot the three curves on the same figure and briefly describe the differences.

4. **Interpretation.** In one page or less, explain why supernova data alone do not directly determine both H_0 and the absolute magnitude M without additional calibration information.

References

[1] A. G. Riess *et al.* [Supernova Search Team], "Observational evidence from supernovae for an accelerating universe and a cosmological constant," *Astron. J.* **116**, 1009 (1998), doi:10.1086/300499, [astro-ph/9805201].

[2] S. Perlmutter *et al.* [Supernova Cosmology Project Collaboration], "Measurements of Omega and Lambda from 42 high redshift supernovae," *Astrophys. J.* **517**, 565 (1999), doi:10.1086/307221, [astro-ph/9812133].

[3] W. H. Press, S. A. Teukolsky, W. T. Vetterling and B. P. Flannery, "Numerical Recipes in FORTRAN: The Art of Scientific Computing."